

Sami Hadouaj

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SUMMARY

PhD Researcher building scalable reasoning systems that bring probabilistic inference to large-scale ML pipelines. My research transforms structured models into optimized inference algorithms, matching the performance of hand-tuned implementations at scale. Developed end-to-end in C++ with JIT compilation and parallel execution, this approach is validated on datasets exceeding 1 billion tokens. My work sits at the intersection of ML, structured reasoning, and high-performance systems, with direct relevance to scalable inference and structured reasoning in modern AI, including scientific and foundation model settings.

EDUCATION

University of Michigan (Rackham Graduate School)	Dearborn, Michigan
Ph.D in Computer and Information Science, GPA: 3.96/4, Advised by: Dr. Niccolò Meneghetti	Jan 2022 – 2026
Tunisia Polytechnic School	Tunis, Tunisia
Master of Science in Computer Science	Sep 2020 – Jun 2021
National Institute of Applied Science and Technology	Tunis, Tunisia
Bachelor in Software Engineering	Sep 2016 – Jun 2021

RESEARCH EXPERIENCE

Graduate Research Assistant	Jan 2022 – Present
University of Michigan-Dearborn	Dearborn, Michigan
- Designed an automated compilation framework that synthesizes optimized Stochastic Variational Inference (SVI) algorithms from structured probabilistic model specifications, enabling scalable approximate Bayesian inference without manual algorithm engineering. - Applied knowledge compilation techniques to eliminate redundant latent variables from probabilistic models, reducing the effective parameter space and accelerating convergence of variational inference. - Benchmarked the framework on a Wikipedia corpus of over 1 billion tokens, demonstrating that the automatically synthesized inference algorithms achieve performance on par with specialized, hand-tuned implementations. - Engineered a general-purpose probabilistic programming system in C++, integrating Apache Arrow for columnar in-memory data management, ClangJIT for just-in-time compilation of inference kernels, and OpenMP for multi-threaded parallel execution.	
Undergraduate Research Assistant	May 2020 – May 2021
University of Michigan-Dearborn	Remote
Engineered a recommendation engine that mapped bug reports to actionable code fixes, streamlining the debugging lifecycle for developers. Advised by Dr. Marouane Kessentini.	

PUBLICATIONS & TECHNICAL REPORTS

- **Sami Hadouaj**, Ouael Ben Amara, Niccolò Meneghetti. “Variational Inference for De Finetti Logic” *Proceedings of the ACM on Management of Data* (**SIGMOD 2026**, to appear).
- Ouael Ben Amara^{*}, **Sami Hadouaj**^{*}, Niccolò Meneghetti. “StarfishDB: a query execution engine for relational probabilistic programming” ¹ *Proceedings of the ACM on Management of Data* 2.3 (2024): 1-31. doi:10.1145/3654988 (**SIGMOD 2024**)
[Paper] [Code] [Poster] [Slides]
- **Sami Hadouaj**, Xiaojun Dong. “Reproducibility report for ACM SIGMOD 2024 paper: ‘On efficient large sparse matrix chain multiplication’ ” *Reproducibility Reports of the 2024 International Conference on Management of Data*. 2025. doi:10.1145/3687998.3717039 (**SIGMOD 2024 Reproducibility**)

¹Sami Hadouaj and Ouael Ben Amara are both first authors of this paper.

- Krause, Alexander, Georgiy Lebedev, **Sami Hadouaj**, Ouael Ben Amara, and Niccolò Meneghetti
“Reproducibility report for ACM SIGMOD 2024 paper: ‘StarfishDB: a query execution engine for relational probabilistic programming’ ” *Reproducibility Reports of the 2024 International Conference on Management of Data*. 2025. doi:10.1145/3687998.3717056 (**SIGMOD 2024 Reproducibility**)

SKILLS

Systems Engineering: C++ (primary), JIT compilation (ClangJIT), parallel computing (OpenMP), columnar data systems (Apache Arrow), build systems (CMake, Ninja), profiling & debugging (perf, gdb, Valgrind)

Languages & Tools: Python, SQL, Java, Datalog, Bash | Linux, Git, Docker

Research Areas: Uncertainty Quantification, Probabilistic Programming, Approximate Bayesian Inference (SVI, MCMC), Scalable Inference, Database Systems, Machine Learning, Generative Models

SERVICE

BIGDATA (2022, 2023, 2024, 2025): External reviewer

ICDM (2025): External reviewer

SIGMOD ARI (2024): Artifacts reviewer: Conducted independent code evaluation and experiment reproduction.

Volunteered with Accelerate4kids, developing educational materials to expand learning access for underprivileged children. (Sep - Dec 2024)

TEACHING

Teaching Assistant, CIS 536: Text Mining & Information Retrieval (with Dr. Niccolò Meneghetti) F2026

Lab Instructor, CIS 200: Software Engineering F2025

Teaching Assistant, CIS 4951: Senior Design I and II (with Dr. Bruce Maxim) Summer 2025

Teaching Assistant, CIS 556: Database Systems (with Dr. Niccolò Meneghetti) F2023, W2024, F2024, W2025

HONORS & AWARDS

Rackham Doctoral Intern Fellowship Program: Selected and funded by the University of Michigan Rackham Graduate School to conduct research during Summer 2024.

TALKS & PRESENTATIONS

SIGMOD 2024 Conference: *StarfishDB: A Query Execution Engine for Relational Probabilistic Programming*
Paper Presentation · Santiago, Chile · June 2024

Northeast Database Day 2026: *Stochastic Variational Inference for Relational Probabilistic Programming*
Poster Presentation · Boston, MA · January 2026

Northeast Database Day 2025: *Declarative Probabilistic Programming for Explainable AI*
Poster Presentation · Boston, MA · January 2025

SIGMOD 2024 Conference: *StarfishDB: A Query Execution Engine for Relational Probabilistic Programming*
Poster Presentation · Santiago, Chile · June 2024

Northeast Database Day 2024: *StarfishDB: A Query Execution Engine for Relational Probabilistic Programming*
Poster Presentation · Boston, MA · March 2024

Northeast Database Day 2023: *Query-driven Inference in Probabilistic Databases*
Poster Presentation · Boston, MA · March 2023

HOBBIES

Running; Cycling; Rock Climbing